Header Attribute	Details
Date	19 October 2026
Team ID	
Project Name	Health360: Clinical & Healthcare Risk Analysis
Maximum Marks	4 Marks

Project Design Phase-I: Solution Architecture

Solution Architecture

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram (Layered Mapping for Health360)

1. **DATA SOURCES:** Structured Clinical CSV (`smart_healthcare_dataset.csv`) containing 5,000 patient vitals and demographic attributes.
2. **DATA INGESTION / CONNECTION:** Tableau Native Connectors, Python Data Loaders, and File Ingestion pipelines.

3. **DATA PREPARATION / PROCESSING:** Data cleansing, handling nulls, demographic cohort binning (6 age tiers: Teenager to Senior), normalizing health risk score index.
4. **DATA STORAGE / DATA LAYER:** Clean In-Memory Data Extracts (Tableau `.hyper`), Normalized CSV Repository.
5. **ANALYTICS / BI LAYER:** Clinical risk metric aggregations, average systolic BP calculations (~134.4 mmHg), fatigue incidence (50.8%), exercise vs. risk correlations.
6. **REPORTING / VISUALIZATION:** 6 interactive clinical charts (Bar, Pie, Stacked Bar, Donut, Substance Bar, Heatmap) and a 5-point executive clinical data story.
7. **BUSINESS USERS:** Clinical Risk Analysts, Healthcare Clinicians, Public Health Officers, and Healthcare Administrators.

Component Description

Component Name	Description / Role in Architecture	Technology Used
Data Sources Layer	Ingests baseline clinical records of 5,000 patients across 6 age groups, capturing systolic blood pressure, fatigue, exercise tiers, and substance use.	Structured CSV (`smart_healthcare_dataset.csv`) / Patient Records
Ingestion & Transformation Layer	Cleans raw metrics, categorizes demographic cohorts, calculates vital thresholds, and validates normalized health risk scoring (Mean 100.0).	Python (Pandas), Tableau Data Engine
Analytical & BI Storage Layer	Manages aggregated clinical data cubes and in-memory extracts for ultra-fast query performance and zero-lag filtering across 5,000 patient records.	Tableau Hyper Data Engine / In-Memory Storage

Component Name	Description / Role in Architecture	Technology Used
Visualization & Reporting Layer	Renders 6 core interactive diagnostic charts and an executive 5-point data story presentation for clinical risk exploration.	Tableau Public / Desktop
Web Presentation Layer	Delivers a responsive, public web dashboard with KPI cards and interactive Chart.js visualizations for cross-device access.	GitHub Pages, HTML5, CSS3, JavaScript (Chart.js)
Business User Consumption Layer	Serves interactive clinical intelligence to end-users for identifying high-risk cohorts and deploying early lifestyle interventions.	Web Browsers (Chrome, Edge), Tableau Public Cloud Viewer